A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.2.1: Multimedia: Graphics

Concept 1.2.1.2: Bitmap File Size Calculation

Total Questions: 8

Mark Schemes begin on Page 6

Question 2 | Source: 9618_w25_qp_11 (Q7.d.i)

7 Web browsers use Internet Protocol (IP) addresses.

(d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

(i) Calculate an estimate of the file size of the bitmap in kibibytes.

Show your working.

Working ....

....

....

....

File size of bitmap .... kibibytes

[2]

Question 3 | Source: 9618_w25_qp_12 (Q6.d)

(d) A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

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File size .... megabytes

[2]

Question 4 | Source: 9618_s25_qp_13 (Q1.a.i)

1 **(a)** A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

(i) Calculate the file size of the image created by the digital camera in megabytes (MB).

Show your working.

Working space ....

....

....

Answer .... MB

[2]

Question 5 | Source: 9618_s24_qp_13 (Q2.a)

2 A photograph is stored as a bitmap image.

(a) The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes.

Calculate an estimate for the file size of the photograph in megabytes.

Show your working.

Working ....

....

....

Answer .... megabytes

[2]

Question 6 | Source: 9618_w23_qp_12 (Q6.c)

(c) A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working ....

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....

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....

....

Estimated file size in mebibytes ....

[2]

Question 7 | Source: 9618_w23_qp_13 (Q2.b)

(b) Calculate the file size of a bitmap image using the following information:

- image resolution of 2048 pixels wide and 1024 pixels high
- bit depth of 16 bits.

Give your answer in kibibytes. Show your working.

Working ....

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....

Answer in kibibytes ....
[2]

Question 8 | Source: 9618_s23_qp_11 (Q1.b.ii)

1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

(b) The bitmap images are photographs of the holiday locations.

(ii) One of the photographs has a bit depth of 8 bytes and an image resolution of 1500 pixels wide and 3000 pixels high.

Calculate the file size of the photograph in megabytes. Show your working.

Working ....

....

....

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File size .... MB

[2]

Question 9 | Source: 9618_s21_qp_11 (Q1.a.ii)

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(ii) The image is scanned with an image resolution of 1024 × 512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working ....

....

....

....

Answer .... mebibytes

[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q7.b)

7(b)	**1 mark** for the working **1 mark** for the correct answer Working: (4000*3000*30*16) / (8*1000*1000*1000) //(4000*3000*30*2) / (1000*1000*1000) Answer: 0.72 gigabytes	**2**

Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q7.d.i)

7(d)(i)	**1 mark** for the working
	1 mark for the correct answer
	
	Working:
	(512 * 2048 * 8) / (8 * 1024) // (512 * 2048) / 1024
	// (2^9 * 2^{11} * 2^3) / (2^3 * 2^{10}) // (2^9 * 2^{11}) / 2^{10}
	
	Answer: 1024 kibibytes // 2^{10} kibibytes
		2

Mark Scheme for Question 3 | Source: 9618_w25_ms_12 (Q6.d)

6(d)	**1 mark** for the working **1 mark** for the correct answer Working: (1000 * 2000 * 16) / (8 * 1000 * 1000) // (1000 * 2000 * 2 / (1000 * 1000) Answer: 4 megabytes	**2**

Mark Scheme for Question 4 | Source: 9618_s25_ms_13 (Q1.a.i)

1(a)(i)	**1 mark** for correct working: e.g. 2000000 * 16 / (8 * 1000 * 1000) **1 mark** for answer: 4MB	**2**

Mark Scheme for Question 5 | Source: 9618_s24_ms_13 (Q2.a)

2(a)	**1 mark** for working: • 4000 * 3000 * 4 **1 mark** for correct answer: • 48MB	**2**

Mark Scheme for Question 6 | Source: 9618_w23_ms_12 (Q6.c)

6(c)	**1 mark** for answer, **1 mark** for working Answer: 2.5 mebibytes Working: (2048 × 1024 × 10) / (8 × 1024 × 1024)	**2**

Mark Scheme for Question 7 | Source: 9618_w23_ms_13 (Q2.b)

2(b)	**1 mark** for answer. **1 mark** for working Answer: 4096 kibibytes Working: File size = (2048 × 1024 × 16) / (8 × 1024)	**2**

Mark Scheme for Question 8 | Source: 9618_s23_ms_11 (Q1.b.ii)

1(b)(ii)	**1 mark** for working; **1 mark** for answer • Working: (1500 * 3000 * 8) / 1000 / 1000 • Answer: 36 MB	**2**

Mark Scheme for Question 9 | Source: 9618_s21_ms_11 (Q1.a.ii)

1(a)(ii)	**1 mark** per bullet point for working, **1 mark** for answer Working: • 1024 × 512 = 524 288 pixels/bytes • 524288 / 1024 / 1024 Answer: 0.50 mebibytes	**3**